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## Gate structures for semiconductor devices


### Abstract

A semiconductor device with different gate structure configurations and a method of fabricating the same are disclosed. The semiconductor device includes first and second pair of source/drain regions disposed on a substrate, first and second nanostructured channel regions, and first and second gate structures with effective work function values different from each other. The first and second gate structures include first and second high-K gate dielectric layers, first and second barrier metal layers with thicknesses different from each, first and second work function metal (WFM) oxide layers with thicknesses substantially equal to each other disposed on the first and second barrier metal layers, respectively, a first dipole layer disposed between the first WFM oxide layer and the first barrier metal layer, and a second dipole layer disposed between the second WFM oxide layer and the second barrier metal layer.



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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/837,859, titled “Gate Structures for Semiconductor Devices,” filed Jun. 10, 2022, which is a continuation of U.S. patent application Ser. No. 16/835,916, titled “Gate Structures for Semiconductor Devices,” filed Mar. 31, 2020, which claims the benefit of U.S. Provisional Patent Application No. 62/928,557, titled “Gate Structures for Threshold Voltage Tuning of FinFETs and Gate All Around FETs,” filed Oct. 31, 2019, each of which is incorporated by reference herein in its entirety.

BACKGROUND
(1) With advances in semiconductor technology, there has been increasing demand for higher storage capacity, faster processing systems, higher performance, and lower costs. To meet these demands, the semiconductor industry continues to scale down the dimensions of semiconductor devices, such as metal oxide semiconductor field effect transistors (MOSFETs), including planar MOSFETs and fin field effect transistors (finFETs). Such scaling down has increased the complexity of semiconductor manufacturing processes.

## Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Aspects of this disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the common practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may

be arbitrarily increased or reduced for clarity of discussion.
(2) FIGS. **1**A, **1**B-**1**E, and **1**F-**1**O illustrate an isometric view, cross-sectional views, and device characteristics of a semiconductor device with different gate structures, in accordance with some embodiments.
(3) FIG. **2** is a flow diagram of a method for fabricating a semiconductor device with different gate structures, in accordance with some embodiments.
(4) FIGS. **3**A-**14**B illustrate cross-sectional views of a semiconductor device with different gate structures at various stages of its fabrication process, in accordance with some embodiments.
(5) Illustrative embodiments will now be described with reference to the accompanying drawings. In the drawings, like reference numerals generally indicate identical, functionally similar, and/or structurally similar elements.

DETAILED DESCRIPTION

(6) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the process for forming a first feature over a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. As used herein, the formation of a first feature on a second feature means the first feature is formed in direct contact with the second feature. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition does not in itself dictate a relationship between the various embodiments and/or configurations discussed.
(7) Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.
(8) It is noted that references in the specification to “one embodiment,” “an embodiment,” “an example embodiment,” “exemplary,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases do not necessarily refer to the same embodiment. Further, when a particular feature, structure or characteristic is described in connection with an embodiment, it would be within the knowledge of one skilled in the art to effect such feature, structure or characteristic in connection with other embodiments whether or not explicitly described.
(9) It is to be understood that the phraseology or terminology herein is for the purpose of description and not of limitation, such that the terminology or phraseology of the present specification is to be interpreted by those skilled in relevant art(s) in light of the teachings herein.
(10) As used herein, the term “etch selectivity” refers to the ratio of the etch rates of two different materials under the same etching conditions.
(11) As used herein, the term “high-k” refers to a high dielectric constant. In the field of semiconductor device structures and manufacturing processes, high-k refers to a dielectric constant that is greater than the dielectric constant of SiO.sub.2 (e.g., greater than 3.9).
(12) As used herein, the term “p-type” defines a structure, layer, and/or region as being doped with p-type dopants, such as boron.
(13) As used herein, the term “n-type” defines a structure, layer, and/or region as being doped with

n-type dopants, such as phosphorus.
(14) As used herein, the term “nanostructured” defines a structure, layer, and/or region as having a horizontal dimension (e.g., along an X- and/or Y-axis) and/or a vertical dimension (e.g., along a Z-axis) less than, for example, 100 nm.
(15) As used herein, the term “n-type work function metal (nWFM)” defines a metal or a metal-containing material with a work function value closer to a conduction band energy than a valence band energy of a material of a FET channel region. In some embodiments, the term “n-type work function metal (nWFM)” defines a metal or a metal-containing material with a work function value less than 4.5 eV.
(16) As used herein, the term “p-type work function metal (pWFM)” defines a metal or a metal-containing material with a work function value closer to a valence band energy than a conduction band energy of a material of a FET channel region. In some embodiments, the term “p-type work function metal (pWFM)” defines a metal or a metal-containing material with a work function value equal to or greater than 4.5 eV.
(17) In some embodiments, the terms “about” and “substantially” can indicate a value of a given quantity that varies within 5% of the value (e.g., ±1%, ±2%, ±3%, ±4%, ±5% of the value).
(18) The fin structures disclosed herein may be patterned by any suitable method. For example, the fin structures may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in some embodiments, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the fin structures.
(19) The required gate voltage—the threshold voltage Vt—to turn on a field effect transistor (FET) can depend on the semiconductor material of the FET channel region and/or the effective work function (EWF) value of a gate structure of the FET. For example, for an n-type FET (NFET), reducing the difference between the EWF value(s) of the NFET gate structure and the conduction band energy of the material (e.g., 4.1 eV for Si or 3.8 eV for SiGe) of the NFET channel region can reduce the NFET threshold voltage. For a p-type FET (PFET), reducing the difference between the EWF value(s) of the PFET gate structure and the valence band energy of the material (e.g., 5.2 eV for Si or 4.8 eV for SiGe) of the PFET channel region can reduce the PFET threshold voltage. The EWF values of the FET gate structures can depend on the thickness and/or material composition of each of the layers of the FET gate structure. As such, FETs can be manufactured with different threshold voltages by adjusting the thickness and/or material composition of the FET gate structures.
(20) Due to the increasing demand for multi-functional portable devices, there is an increasing demand for FETs with different threshold voltages on the same substrate. One way to achieve such FETs can be with different work function metal (WFM) layer thicknesses in the FET gate structures. However, the different WFM layer thicknesses can be constrained by the FET gate structure geometries. For example, in gate-all-around (GAA) FETs, the WFM layer thicknesses can be constrained by the spacing between the nanostructured channel regions of the GAA FETs. Also, depositing different WFM layer thicknesses can become increasingly challenging with the continuous scaling down of FETs (e.g., GAA FETs and/or finFETs).
(21) The present disclosure provides example FET gate structures with different EWF values to form FETs (e.g., GAA FETs and/or finFETs) with different threshold voltages and provides example methods of forming such FETs on a same substrate. The example methods form NFETs and PFETs with similar WFM layer thickness, but with different threshold voltages on the same substrate. These example methods can be less complicated and more cost-effective in

manufacturing reliable gate structures with lower gate resistance in FETs with nanostructured channel regions and with different threshold voltages than other methods of forming FETs with similar channel dimensions and threshold voltages on the same substrate. In addition, these example methods can form FET gate structures with smaller dimensions (e.g., thinner gate stacks) than other methods of forming FETs with similar threshold voltages.

(22) In some embodiments, NFETs and PFETs with different gate structure configurations, but with similar WFM layer thicknesses can be selectively formed on the same substrate to achieve threshold voltages different from each other. The different gate structure configurations can have barrier metal layers of different thicknesses disposed between the WFM layers and high-K gate dielectric layers. In addition, the WFM layers can include WFM oxide layers that induce dipole layers at the interface between the WFM layers and the barrier metal layers. The different barrier metal layer thicknesses provide different spacings between the WFM layers and high-K gate dielectric layers and different spacings between the induced dipole layers and the high-K gate dielectric layers. These different spacings result in the FET gate structures having EWF values different from each other and consequently having threshold voltages different from each other. Thus, tuning the barrier metal layer thicknesses can tune the EWF values of the NFET and PFET gate structures and, as a result, adjust the threshold voltages of the NFETs and PFETs without varying their WFM layer thicknesses.

(23) A semiconductor device **100** having NFETs **102**N**1**-**102**N**3** and PFETs **102**P**1**-**102**P**3** is described with reference to FIGS. **1**A-**1**O, according to some embodiments. FIG. **1**A illustrates an isometric view of semiconductor device **100**, according to some embodiments. FIGS. **1**B-**1**C and **1**D-**1**E illustrate cross-sectional views along lines A-A and B-B of semiconductor device **100** of FIG. **1**A, according to some embodiments. FIGS. **1**F-**1**O illustrate devices characteristics of semiconductor device **100**, according to some embodiments. Even though six FETs are discussed with reference to FIGS. **1**A-**1**O, semiconductor device **100** can have any number of FETs. The discussion of elements of NFETs **102**N**1**-**102**N**3** and PFETs **102**P**1**-**102**P**3** with the same annotations applies to each other, unless mentioned otherwise. The isometric view and cross-sectional views of semiconductor device **100** are shown for illustration purposes and may not be drawn to scale.

(24) Referring to FIGS. **1**A-**1**C, NFETs **102**N**1**-**102**N**3** and PFETs **102**P**1**-**102**P**3** can be formed on a substrate **106**. Substrate **106** can be a semiconductor material such as, but not limited to, silicon. In some embodiments, substrate **106** can include a crystalline silicon substrate (e.g., wafer). In some embodiments, substrate **106** can include (i) an elementary semiconductor (e.g., germanium (Ge)); (ii) a compound semiconductor including II-v semiconductor material; (iii) an alloy semiconductor (e.g., silicon germanium (SiGe)); (iv) a silicon-on-insulator (SOI) structure; (v) a silicon germanium (SiGe)-on insulator structure (SiGeOI); (vi) germanium-on-insulator (GeOI) structure; or (vii) a combination thereof. Further, substrate **106** can be doped with p-type dopants (e.g., boron, indium, aluminum, or gallium) or n-type dopants (e.g., phosphorus or arsenic).

(25) NFETs **102**N**1**-**102**N**3** and PFETs **102**P**1**-**102**P**3** can include fin structures **108**.sub.1-**108**.sub.2 extending along an X-axis, epitaxial fin regions **110**A-**110**B, gate structures **112**N**1**-**112**N**3** and **112**P**1**-**112**P**3** inner spacers **142**, and gate spacers **114**.

(26) Referring to FIGS. **1**B-**1**C, fin structure **108**.sub.1 can include a fin base portion **108**A and nanostructured channel regions **120**N disposed on fin base portion **108**A and fin structure **108**.sub.2 can include a fin base portion **108**B and nanostructured channel regions **120**P disposed on fin base portion **108**B. In some embodiments, fin base portions **108**A-**108**B can include a material similar to substrate **106**. Nanostructured channel regions **120**N can be wrapped around by gate structures **112**N**1**-**112**N**3** and nanostructured channel regions **120**P can be wrapped around by gate structures **112**P**1**-**112**P**3**. Nanostructured channel regions **120**N-**120**P can include semiconductor materials similar to or different from substrate **106** and can include semiconductor material similar to or different from each other.

(27) In some embodiments, nanostructured channel regions **120**N can include Si, SiAs, silicon phosphide (SiP), SiC, or silicon carbon phosphide (SiCP) for NFETs **102**N**1**-**102**N**3** and nanostructured channel regions **120**P can include SiGe, silicon germanium boron (SiGeB), germanium boron (GeB), silicon germanium stannum boron (SiGeSnB), or a III-V semiconductor compound for PFETs **102**P**1**-**102**P**3**. In some embodiments, nanostructured channel regions **120**N-**120**P can both include Si, SiAs, SiP, SiC, SiCP, SiGe, SiGeB, GeB, SiGeSnB, or a III-V semiconductor compound. Though rectangular cross-sections of nanostructured channel regions **120**N-**120**P are shown, nanostructured channel regions **120**N-**120**P can have cross-sections of other geometric shapes (e.g., circular, elliptical, triangular, or polygonal).

(28) Referring to FIGS. **1**A-**1**C, epitaxial fin regions **110**A-**110**B can be grown on fin base portions **108**A-**108**B, respectively, and can be source/drain (S/D) regions of NFETs **102**N**1**-**102**N**3** and PFETs **102**P**1**-**102**P**3**. Epitaxial fin regions **110**A-**110**B can include epitaxially-grown semiconductor materials similar to or different from each other. In some embodiments, the epitaxially-grown semiconductor material can include the same material or a different material from the material of substrate **106**. Epitaxial fin regions **110**A and **110**B can be n- and p-type, respectively. In some embodiments, n-type epitaxial fin regions **110**A can include SiAs, SiC, or SiCP. P-type epitaxial fin regions **110**B can include SiGe, SiGeB, GeB, SiGeSnB, a III-V semiconductor compound, or a combination thereof.

(29) Gate structures **112**N**1**-**112**N**3** and **112**P**1**-**112**P**3** can be multi-layered structures. Gate structures **112**N**1**-**112**N**3** can be wrapped around nanostructured channel regions **120**N and gate structures **112**P**1**-**112**P**3** can be wrapped around nanostructured channel regions **120**P for which gate structures **112**N**1**-**112**N**3** and **112**P**1**-**112**P**3** can be referred to as “gate-all-around (GAA) structures” or “horizontal gate-all-around (HGAA) structures.” NFETs **102**N**1**-**102**N**3** and PFETs **102**P**1**-**102**P**3** can be referred to as “GAA FETs **102**N**1**-**102**N**3** and **102**P**1**-**102**P**3**” or “GAA NFETs **102**N**1**-**102**N**3** and PFETs **102**P**1**-**102**P**3**,” respectively.

(30) In some embodiments, NFETs **102**N**1**-**102**N**3** and PFETs **102**P**1**-**102**P**3** can be finFETs and have fin regions **120**N*-**120**P* instead of nanostructures channel regions **120**N-**120**P, as shown in FIGS. **1**D-**1**E. Such finFETs **102**N**1**-**102**N**3** and **102**P**1**-**102**P**3** can have gate structures **112**N**1**-**112**N**3** and **112**P**1**-**112**P**3** disposed on fin regions **120**N*-**120**P* as shown in FIGS. **1**D-**1**E, respectively.

(31) Gate structures **112**N**1**-**112**N**3** and **112**P**1**-**112**P**3** can include interfacial oxide layers **127**, high-k (HK) gate dielectric layers **128**N**1**-**128**N**3** and **128**P**1**-**128**P**3**, first barrier metal layers **129**N**1**-**129**N**3** and **129**P**1**-**129**P**3**, barrier metal oxide layers **130**N**1**-**130**N**3** and **130**P**1**-**130**P**3**, dipole layers **131**N**1**-**131**N**3** and **131**P**1**-**131**P**3**, WFM oxide layers **132**N**1**-**132**N**3** and **132**P**1**-**132**P**3**, second barrier metal layers **133**, fluorine-free tungsten (FFW) layers **134**, and gate metal fill layers **135**. Even though FIGS. **1**B-**1**C show that all the layers of gate structures **112**N**1**-**112**N**3** and **112**P**1**-**112**P**3** are wrapped around nanostructured channel regions **120**N-**120**P, nanostructured channel regions **120**N-**120**P can be wrapped around by at least interfacial oxide layers **127** and HK gate dielectric layers **128**N**1**-**128**N**3** and **128**P**1**-**128**P**3** to fill the spaces between adjacent nanostructured channel regions **120**N-**120**P. As such, nanostructured channel regions **120**N can be electrically isolated from each other to prevent shorting between gate structures **112**N**1**-**112**N**3** and S/D regions **110**A during operation of NFETs **102**N**1**-**102**N**3**. Similarly, nanostructured channel regions **120**P can be electrically isolated from each other to prevent shorting between gate structures **112**P**1**-**112**P**3** and S/D regions **110**B during operation of PFETs **102**P**1**-**102**P**3**.

(32) Interfacial oxide layers **127** can be disposed on nanostructured channel regions **120**N-**120**P and can include silicon oxide and a thickness ranging from about 0.5 nm to about 1.5 nm. Each of HK gate dielectric layers **128**N**1**-**128**N**3** and **128**P**1**-**128**P**3** can have a thickness (e.g., about 1 nm to about 3 nm) that is about 2 to 3 times the thickness of interfacial oxide layers **127** and can include (i) a high-k dielectric material, such as hafnium oxide (HfO.sub.2), titanium oxide (TiO.sub.2), hafnium zirconium oxide (HfZrO), tantalum oxide (Ta.sub.2O.sub.3), hafnium silicate

(HfSiO.sub.4), zirconium oxide (ZrO.sub.2), and zirconium silicate (ZrSiO.sub.2) and (ii) a high-k dielectric material having oxides of lithium (Li), beryllium (Be), magnesium (Mg), calcium (Ca), strontium (Sr), scandium (Sc), yttrium (Y), zirconium (Zr), aluminum (Al), lanthanum (La), cerium (Ce), praseodymium (Pr), neodymium (Nd), samarium (Sm), europium (Eu), gadolinium (Gd), terbium (Tb), dysprosium (Dy), holmium (Ho), erbium (Er), thulium (Tm), ytterbium (Yb), lutetium (Lu), or (iii) a combination thereof.

(33) First barrier metal layers **129**N**1**-**129**N**3** and **129**P**1**-**129**P**3** can be disposed on and in physical contact with HK gate dielectric layers **128**N**1**-**128**N**3** and **128**P**1**-**128**P**3**, respectively. In some embodiments, first barrier metal layers **129**N**1**-**129**N**3** and **129**P**1**-**129**P**3** can include metal nitrides (e.g., TiN and/or TaN) or any material that can prevent material diffusion from overlying layers (e.g., WFM oxide layers **132**N**1**-**132**N**3** and **132**P**1**-**132**P**3**) to HK gate dielectric layers **128**N**1**-**128**N**3** and **128**P**1**-**128**P**3**. Each first barrier metal layers **129**N**1**-**129**N**3** and **129**P**1**-**129**P**3** can include a single layer of metal nitride or a stack of metal nitride layers. The stack of metal nitride layers can include two or more metal nitride layers that are similar to or different from each other.

(34) Referring to FIG. **1**B, thicknesses T**1**-T**3** of first barrier metal layers **129**N**1**-**129**N**3** can be different from each other to provide different spacings S**1**-S**3** between WFM metal oxide layers **132**N**1**-**132**N**3** and HK gate dielectric layers **128**N**1**-**128**N**3**, respectively. The different thicknesses T**1**-T**3** can also provide different spacings S**4**-S**6** between dipole layers **131**N**1**-**131**N**3** and HK gate dielectric layers **128**N**1**-**128**N**3**. In some embodiments, thickness T**3** is greater than thickness T**2**, which is greater than thickness T**1**. As a result, spacings S**3** and S**6** can be greater than spacings S**2** and S**5**, which can be greater than spacings Si and S**4**, respectively. The different spacings S**1**-S**3** and/or S**4**-S**6** can result in gate structures **112**N**1**-**112**N**3** having EWF values E**1**-E**3** different from each other and consequently having threshold voltages V**1**-V**3** different from each other. Thicknesses T**1**-T**3** and S**4**-S**6** can be directly proportional to EWF values E**1**-E**3** and threshold voltages V**1**-V**3** as shown in FIGS. **1**F-**1**I.

(35) Similarly, referring to FIG. **1**C, thicknesses T**4**-T**6** of first barrier metal layers **129**P**1**-**129**P**3** can be different from each other to provide different spacings S**7**-S**9** between WFM metal oxide layers **132**P**1**-**132**P**3** and HK gate dielectric layers **128**P**1**-**128**P**3**. The different thicknesses T**4**-T**6** can also provide different spacings S**10**-S**12** between dipole layers **131**P**1**-**131**P**3** and HK gate dielectric layers **128**P**1**-**128**P**3**. Thickness T**6** is greater than thickness T**5**, which is greater than thickness T**4**. As a result, spacings S**7** and S**10** can be greater than spacings S**8** and S**11**, which can be greater than spacings S**9** and S**12**, respectively. The different spacings S**7**-S**9** and/or S**10**-S**12** can result in gate structures **112**P**1**-**112**P**3** having EWF values E**4**-E**6** different from each other and consequently having threshold voltages V**4**-V**6** different from each other. Thicknesses T**4**-T**6** and S**10**-S**12** can be directly proportional to EWF values E**4**-E**6** and indirectly proportional to threshold voltages V**4**-V**6** as shown in FIGS. **1**J-**1**M.

(36) In some embodiments, thicknesses T**1**-T**3** can be similar to or different from thicknesses T**4**-T**6**, respectively. Even with thicknesses T**1**-T**3** similar to respective thicknesses T**4**-T**6**, threshold voltages V**1**-V**3** can be different from threshold voltages V**4**-V**6**, respectively. In some embodiments, thicknesses T**1**-T**6** can range from about 0.5 nm to about 3 nm.

(37) Barrier metal oxide layers **130**N**1**-**130**N**3** and **130**P**1**-**130**P**3** can be disposed on first barrier metal layers **129**N**1**-**129**N**3** and **129**P**1**-**129**P**3**, respectively. In some embodiments, barrier metal oxide layers **130**N**1**-**130**N**3** and **130**P**1**-**130**P**3** can include an oxide of the metal included in metal nitrides of first barrier metal layers **129**N**1**-**129**N**3** and **129**P**1**-**129**P**3**. For example, barrier metal oxide layers **130**N**1**-**130**N**3** and **130**P**1**-**130**P**3** can include an oxide of Ti (e.g., TiO.sub.x) or Ta (e.g., TaO.sub.x) when TiN or TaN is included in first barrier metal layers **129**N**1**-**129**N**3** and **129**P**1**-**129**P**3**. In some embodiments, thicknesses of each barrier metal oxide layers **130**N**1**-**130**N**3** and **130**P**1**-**130**P**3** can range from about 0.1 nm to about 0.2 nm. Barrier metal oxide layers **130**P**1**-**130**P**3** are thicker than barrier metal oxide layers **130**N**1**-**130**N**3** as a result of first barrier metal layers **129**P**1**-**129**P**3** being oxidized more times than first barrier metal layers **129**N**1**-**129**N**3** during

the fabrication of gate structures 112N1-112N3 and 112P1-112P3 as described below with reference to FIGS. 3A-14B.

(38) WFM oxide layers 132N1-132N3 and 132P1-132P3 can be disposed on and in physical contact with barrier metal oxide layers 130N1-130N3 and 130P1-130P3, respectively. For NFETs 102N1-102N3, n-type WFM oxide layers 132N1-132N3 (also referred to as “nWFM oxide layers 132N1-132N3”) can include oxides of Al-free (e.g., with no Al) metals. In some embodiments, WFM oxide layers 132N1-132N3 can include (i) rare-earth metal (REM) oxides, such as lanthanum oxide (La.sub.2O.sub.3), cerium oxide (CeO.sub.2), ytterbium oxide (Yb.sub.2O.sub.3), lutetium oxide (Lu.sub.2O.sub.3), and erbium oxide (Er.sub.2O.sub.3); (ii) oxides of a metal from group HA (e.g., magnesium oxide (MgO) or strontium oxide (SrO)), group IUB (e.g., yttrium oxide (Y.sub.2O.sub.3)), group IVB (e.g., hafnium oxide (HfO.sub.2) or zirconium oxide (ZrO.sub.2)), or group VB (e.g., tantalum oxide (Ta.sub.2O.sub.5)) of the periodic table; (iii) silicon dioxide (SiO.sub.2); or (iv) a combination thereof.

(39) In contrast, for PFETs 102P1-102P3, p-type WFM oxide layers 132P1-132P3 132N3 (also referred to as “pWFM oxide layers 132P1-132P3”) can include (i) Al-based metal oxides, such as aluminum oxide (Al.sub.2O.sub.3) and aluminum titanium oxide (Al.sub.2TiO.sub.5); (ii) oxides of a metal from group VB (e.g., niobium oxide (NbO)), group IIA (e.g., boron oxide B.sub.2O.sub.3), group VA (e.g., phosphorus oxide (P.sub.2O.sub.5)) of the periodic table; or (iii) a combination thereof. In some embodiments, thicknesses of each WFM oxide layers 132N1-132N3 and 132P1-132P3 can range from about 0.01 nm to about 2 nm. The thickness within this range can allow WFM oxide layers 132N1-132N3 and 132P1-132P3 to wrap around nanostructured channel regions 120N-120P without being constrained by the spacing between adjacent nanostructured channel regions 120N-120P.

(40) The thickness of WFM oxide layers 132N1-132N3 and 132P1-132P3 can be similar to or different from each other, but the materials of WFM oxide layers 132N1-132N3 are different from the materials of WFM oxide layers 132P1-132P3. In some embodiments, the materials of WFM oxide layers 132N1-132N3 can include a metal oxide with a work function value closer to a conduction band energy than a valence band energy of a material of nanostructured channel regions 120N. In contrast, the materials of WFM oxide layers 132P1-132P3 can include a metal oxide with a work function value closer to a valence band energy than a conduction band energy of a material of nanostructured channel regions 120P.

(41) WFM oxide layers 132N1-132N3 induces dipole layers 131N1-131N3 at the interfaces between WFM oxide layers 132N1-132N3 and barrier metal oxide layers 130N1-130N3. WFM oxide layers 132P1-132P3 induces dipole layers 131P1-131P3 at the interfaces between WFM oxide layers 132P1-132P3 and barrier metal oxide layers 130P1-130P3. Dipole layers 131N1-131N3 and 131P1-131P3 can have dipoles of metal ions and oxygen ions. The metal ions (e.g., La ions) of dipole layers 131N1-131N3 diffuse from the metal oxides (e.g., La.sub.2O.sub.3) of WFM oxide layers 132N1-132N3 and the oxygen ions of dipole layers 131N1-131N3 diffuse from barrier metal oxide layers 130N1-130N3. Similarly, the metal ions (e.g., Al ions) of dipole layers 131P1-131P3 diffuse from the metal oxides (e.g., Al.sub.2O.sub.3) of WFM oxide layers 132P1-132P3 and the oxygen ions of dipole layers 131P1-131P3 diffuse from barrier metal oxide layers 130P1-130P3. The dipoles of dipole layers 131N1-131N3 can have a polarity opposite to a polarity of the dipoles of dipole layers 131P1-131P3. In some embodiments, the concentration of dipoles in dipole layers 131N1-131N3 and 131P1-131P3 can similar to or different from each other.

(42) Second barrier metal layers 133 can be disposed on and in physical contact with WFM oxide layers 132N1-132N3 and 132P1-132P3. In some embodiments, second barrier metal layers 133 can include metal nitrides (e.g., TiN and/or TaN) and can have a thickness ranging from about 1.5 nm to about 3 nm. In some embodiments, the material composition of second barrier metal layers 133 can be similar to the material composition of first barrier metal layers 129N1-129N3 and/or 129P1-129P3.

(43) FFW layers **134** can be disposed on and in physical contact with second barrier metal layers **133**. FFW layers **134** can prevent any substantial diffusion of fluorine (e.g., no fluorine diffusion) from fluorine-based precursors used during the deposition of overlying gate metal fill layers **135** to underlying layers, such as interfacial oxide layers **127**, HK gate dielectric layers **128**N**1**-**128**N**3** and **128**P**1**-**128**P**3**, first barrier metal layers **129**N**1**-**129**N**3** and **129**P**1**-**129**P**3**, WFM oxide layers **132**N**1**-**132**N**3** and **132**P**1**-**132**P**3**, and second barrier metal layers **133**. FFW layers **134** can include substantially fluorine-free tungsten layers. The substantially fluorine-free tungsten layers can include an amount of fluorine contaminants less than about 5 atomic percent in the form of ions, atoms, and/or molecules. In some embodiments, FFW layers **134** can each have a thickness ranging from about 2 nm to about 4 nm for effective blocking of fluorine diffusion from gate metal fill layers **135**.
(44) Gate metal fill layers **135** can each include a single metal layer or a stack of metal layers. The stack of metal layers can include metals different from each other. In some embodiments, gate metal fill layers **135** can include a suitable conductive material, such as W, Ti, silver (Ag), ruthenium (Ru), molybdenum (Mo), copper (Cu), cobalt (Co), Al, iridium (Ir), nickel (Ni), metal alloys, and combinations thereof.
(45) FIG. **1**N illustrates the atomic concentration profiles of oxygen, nitrogen, and lanthanum atoms along lines C of FIG. **1**B when WFM oxide layers **132**N**1**-**132**N**3** include La.sub.2O.sub.3. As shown in FIG. **1**N, the atomic concentration profile of La atoms can have a peak at the interfaces between WFM oxide layers **132**N**1**-**132**N**3** and barrier metal oxide layers **130**N**1**-**130**N**3**, respectively.
(46) FIG. **1**O illustrates the atomic concentration profiles of oxygen, nitrogen, and lanthanum atoms along lines D of FIG. **1**C when WFM oxide layers **132**P**1**-**132**P**3** include Al.sub.2O.sub.3. As shown in FIG. **1**O, the atomic concentration profile of Al atoms can have a peak at the interfaces between WFM oxide layers **132**P**1**-**132**P**3** and barrier metal oxide layers **130**P**1**-**130**P**3**.
(47) Referring back to FIGS. **1**B-**1**E, gate spacers **114** and inner spacers **142** can form sidewalls of gate structures **112**N**1**-**112**N**3** and **112**P**1**-**112**P**3**. Each of gate spacers **114** and/or inner spacers **142** can be in physical contact with interfacial oxide layers **127** and HK gate dielectric layers **128**N**1**-**128**N**3** and **128**P**1**-**128**P**3**, according to some embodiments. Each of gate spacers **114** and inner spacer **142** can include insulating material, such as silicon oxide, silicon nitride, a low-k material, and a combination thereof. Each of gate spacers **114** and inner spacers **142** can have a low-k material with a dielectric constant less than about 3.9.
(48) Semiconductor device **100** can further include etch stop layer (ESL) **116**, interlayer dielectric (ILD) layer **118**, and shallow trench isolation (STI) regions **138**. ESL **116** can be disposed on sidewalls of gate spacers **114** and on epitaxial fin regions **110**A-**110**B. ESL **116** can be configured to protect gate structures **112**N**1**-**112**N**3** and **112**P**1**-**112**P**3** and/or S/D regions **110**A-**110**B. In some embodiments, ESL **116** can include, for example, silicon nitride (SiN.sub.x), silicon oxide (SiO.sub.x), silicon oxynitride (SiON), silicon carbide (SiC), silicon carbo-nitride (SiCN), boron nitride (BN), silicon boron nitride (SiBN), silicon carbon boron nitride (SiCBN), or a combination thereof.
(49) ILD layer **118** can be disposed on ESL **116** and can include a dielectric material deposited using a deposition method suitable for flowable dielectric materials (e.g., flowable silicon oxide, flowable silicon nitride, flowable silicon oxynitride, flowable silicon carbide, or flowable silicon oxycarbide). In some embodiments, the dielectric material is silicon oxide. STI regions **138** can be configured to provide electrical isolation between NFETs **102**N**1**-**102**N**3** and PFETs **102**P**1**-**102**P**3** and neighboring FETs (not shown) on substrate **106** and/or neighboring active and passive elements (not shown) integrated with or deposited on substrate **106**. In some embodiments, STI regions **138** can include silicon oxide, silicon nitride, silicon oxynitride, fluorine-doped silicate glass (FSG), a low-k dielectric material, and/or other suitable insulating materials.
(50) The cross-sectional shapes of semiconductor device **100** and its elements (e.g., fin structure

**108**.sub.1-**108**.sub.2, gate structures **112**N**1**-**112**N**3** and **112**P**1**-**112**P**3**, epitaxial fin regions **110**A-**110**B, inner spacers **142**, gate spacers **114**, and/or STI regions **138**) are illustrative and are not intended to be limiting.

(51) FIG. **2** is a flow diagram of an example method **200** for fabricating semiconductor device **100**, according to some embodiments. For illustrative purposes, the operations illustrated in FIG. **2** will be described with reference to the example fabrication process for fabricating semiconductor device **100** as illustrated in FIGS. **3**A-**14**B. FIGS. **3**A-**14**B are cross-sectional views along lines A-A and B-B of semiconductor device **100** at various stages of fabrication, according to some embodiments. Operations can be performed in a different order or not performed depending on specific applications. It should be noted that method **200** may not produce a complete semiconductor device **100**. Accordingly, it is understood that additional processes can be provided before, during, and after method **200**, and that some other processes may only be briefly described herein. Elements in FIGS. **3**A-**14**B with the same annotations as elements in FIGS. **1**A-**1**O are described above.

(52) In operation **205**, polysilicon structures and epitaxial fin regions are formed on fin structures of NFETs and PFETs. For example, as shown in FIGS. **3**A-**3**B, polysilicon structures **312** can be formed on fin structures **108**.sub.1-**108**.sub.2 and gate spacers **114** can be formed on sidewalls of polysilicon structures **312**. During subsequent processing, polysilicon structures **312** can be replaced in a gate replacement process to form gate structures **112**N**1**-**112**N**3** and **112**P**1**-**112**P**3**. Following the formation of gate spacers **114**, n- and p-type epitaxial fin regions **110**A-**110**B can be selectively formed on portions of fin structures **108**.sub.1-**108**.sub.2 that are not underlying polysilicon structures **312**. After the formation of epitaxial fin regions **110**A-**110**B, ESL **116** and ILD **118** can be formed to form the structures of FIGS. **3**A-**3**B.

(53) Referring to FIG. **2**, in operation **210**, gate openings are formed on and within the one or more fin structures. For example, as shown in FIGS. **4**A-**4**B, gate openings **412**N-**412**P associated with NFETs **102**N**1**-**102**N**3** and PFETs **102**P**1**-**102**P**3**, respectively, can be formed on and within fin structures **108**.sub.1-**108**.sub.2. The formation of gate openings **412**N can include sequential operations of (i) etching polysilicon structures **312** from the structures of FIGS. **3**A-**3**B, and (ii) etching nanostructured regions **122**N-**122**P from the structures of FIGS. **3**A-**3**B. In some embodiments, the etching of nanostructured regions **122**N-**122**P can include using a dry etching process or a wet etching process with higher selectivity towards the material (e.g., SiGe) of nanostructured regions **122**N-**122**P than the material (e.g., Si) of nanostructured channel regions **120**N-**120**P. In some embodiments, the wet etching process can include using a mixture of sulfuric acid (H.sub.2SO.sub.4) and hydrogen peroxide (H.sub.2O.sub.2) (SPM) and/or a mixture of ammonia hydroxide (NH.sub.4OH) with H.sub.2O.sub.2 and deionized (DI) water (APM). In some embodiments, the wet etching process can include using a mixture (NH.sub.4OH) with HCl.

(54) Referring to FIG. **2**, in operations **215**-**230**, gate-all-around (GAA) structures are formed in the gate openings. For example, based on operations **215**-**230**, gate structures **112**N**1**-**112**N**3** and **112**P**1**-**112**P**3** can be wrapped around nanostructured channel regions **120**N-**120**P, as described with reference to FIGS. **5**A-**14**B.

(55) In operation **215**, interfacial oxide layers and an HK gate dielectric layer are deposited and annealed within the gate openings. For example, as shown in FIGS. **5**A-**5**B, interfacial oxide layers **127** and a gate dielectric layer **128** can be deposited and annealed on nanostructured channel regions **120**N-**120**P within gate openings **412**N-**412**P (shown in FIGS. **4**A-**4**B). During subsequent processing, HK gate dielectric layer **128** can form HK gate dielectric layers **128**N**1**-**128**N**3** and **128**P**1**-**138**P**3**, as shown in FIGS. **1**A-**1**E. FIGS. **5**A-**5**B show portions **100**A-**100**B of the structures of FIGS. **4**A-**4**B, respectively, for the sake of clarity.

(56) Interfacial oxide layers **127** can be formed on exposed surfaces of nanostructured channel regions **120**N-**120**P within gate openings **412**N-**412**P, respectively. In some embodiments, interfacial oxide layers **127** can be formed by exposing nanostructured channel regions **120**N-**120**P

to an oxidizing ambient. For example, the oxidizing ambient can include a combination of ozone (O.sub.3), a mixture of ammonia hydroxide, hydrogen peroxide, and water (SC1 solution), and/or a mixture of hydrochloric acid, hydrogen peroxide, water (SC2 solution). As a result of the oxidation process, oxide layers ranging from about 0.5 nm to about 1.5 nm can be formed on the exposed surfaces of nanostructured channel regions **120**N-**120**P.

(57) The deposition of HK gate dielectric layer **128** can include blanket depositing HK gate dielectric layer **128** on the partial semiconductor device **100** (not shown) formed after the formation of interfacial oxide layers **127**. The blanket deposited HK gate dielectric layer **128** can be substantially conformally deposited on interfacial oxide layers **127** and the exposed surfaces of the partial semiconductor device **100** (e.g., sidewalls of gate openings **412**N-**412**P and top surfaces of ILD **118**), as shown in FIGS. **5**A-**5**B. In some embodiments, HK gate dielectric layer **128** can include a dielectric material with a dielectric constant (k-value) higher than about 3.9. In some embodiments, HK gate dielectric layer **128** can include (i) a high-k dielectric material, such as hafnium oxide (HfO.sub.2), TiO.sub.2, HfZrO, Ta.sub.2O.sub.3, HfSiO.sub.4, ZrO.sub.2, and ZrSiO.sub.2, (ii) a high-k dielectric material having oxides of Li, Be, Mg, Ca, Sr, Sc, Y, Zr, Al, La, Ce, Pr, Nd, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, or Lu, or (iii) a combination thereof. In some embodiments, HK gate dielectric layer **128** can be formed with ALD using hafnium chloride (HfCl.sub.4) as a precursor at a temperature ranging from about 250° C. to about 350° C. In some embodiments, gate dielectric layer **128** can have a thickness ranging from about 1 nm to about 3 nm in order to wrap around nanostructures channel regions **120**N-**120**P without being constrained by spacing between adjacent nanostructured channel regions **120**N and between adjacent nanostructured channel regions **120**P.

(58) Referring to FIG. **2**, in operation **220**, first barrier metal layers are formed on the HK gate dielectric layer. For example, FIGS. **6**A-**10**B illustrate the formation of first barrier metal layers **129***a*-**129***c* in one embodiment and FIGS. **11**A-**11**B illustrates the formation of first barrier metal layers **129***a*-**129***c* in another embodiment. During subsequent processing, first barrier metal layers **129***a*-**129***c* can form first barrier metal layers **129**N**1**-**129**N**3** and **129**P**1**-**129**P**3**.

(59) Referring to FIGS. **6**A-**10**B, the formation of first barrier metal layers **129***a*-**129***c* can include sequential operations of (i) blanket depositing a metal nitride layer **129***a** (FIGS. **6**A-**6**B) on the structures of FIGS. **5**A-**5**B, (ii) patterning metal nitride layer **129***a** to selectively form first barrier metal layers **129***a* on NFET **102**N**3** and PFET **102**P**3** as shown in FIGS. **7**A-**7**B, (iii) blanket depositing a metal nitride layer **129***b** (FIGS. **8**A-**8**B) on the structures of FIGS. **7**A-**7**B, (iv) patterning metal nitride layer **129***b** to selectively form first barrier metal layers **129***b* on NFETs **102**N**2**-**102**N**3** and PFETs **102**P**2**-**102**P**3** as shown in FIGS. **9**A-**9**B, and (v) blanket depositing a metal nitride layer for first barrier metal layer **129***c* (FIGS. **10**A-**10**B) on the structures of FIGS. **9**A-**9**B.

(60) Operations (i)-(v) can be repeated to form additional first barrier metal layers similar to first barrier metal layers **129***a*-**129***c* in NFETs **102**N**1**-**102**N**3** and PFETs **102**P**1**-**102**P**3**. In subsequent processing, (i) the portions of first barrier metal layer **129***c* within gate openings **412**N-**412**P of NFET **102**N**1** and PFET **102**P**1** can form first barrier metal layers **129**N**1** and **129**P**1**, respectively, (ii) the portions of first barrier metal layers **129***b*-**129***c* within gate openings **412**N-**412**P of NFET **102**N**2** and PFET **102**P**2** can form first barrier metal layers **129**N**2** and **129**P**2**, respectively, and (iii) the portions of first barrier metal layers **129***a*-**129***c* within gate openings **412**N-**412**P of NFET **102**N**2** and PFET **102**P**2** can form first barrier metal layers **129**N**2** and **129**P**2**, respectively. Thus, the different thicknesses T**1**-T**6** (FIGS. **1**B-**1**C) of first barrier metal layers **129**N**1**-**129**N**3** and **129**P**1**-**129**P**3** can be achieved by selectively forming different number of stacked layers in NFETs **102**N**1**-**102**N**3** and PFETs **102**P**1**-**102**P**3** through multiple deposition and patterning operations.

(61) The blanket deposition of metal nitride layer **129***a** can include blanket depositing about 0.1 nm to about 0.5 nm thick metal nitride layer with an ALD or a CVD process using titanium tetrachloride (TiCl.sub.4) and NH.sub.3 as precursors at a temperature ranging from about 400° C.

to about 450° C. and at a pressure ranging from about 2 torr to about 10 torr. In some embodiments, metal nitride layer **129***a** can be deposited in an ALD process of about 20 cycles to about 30 cycles, where one cycle can include sequential periods of: (i) first precursor gas (e.g., TiCl.sub.4) flow, (ii) a first gas purging process, (iii) a second precursor gas (e.g., NH.sub.3) gas flow, and (iv) a second gas purging process. The blanket deposited metal nitride layer **129***a** can be substantially conformally deposited (e.g., step coverage of about 99%) on the structures of FIGS. **5**A-**5**B. The blanket deposition of metal nitride layers **129***b**-**129***c** can be similar to the blanket deposition of metal nitride layer **129***a**. In some embodiments, the materials of metal nitride layer **129***a****129***c** can include Ti- or Ta-based nitrides or alloys and can be similar to or different from each other.
(62) The patterning of metal nitride layer **129***a****129***c** to form first barrier metal layers **129***a*-**129***c* can include photolithographic and etching processes. The etching process can include a wet etching process using etchants that include a mixture of ammonia hydroxide, hydrogen peroxide, and water (SC1 solution), and/or a mixture of hydrochloric acid, hydrogen peroxide, and water (SC2 solution).
(63) Instead of the different number of stacked layers, the different thicknesses T**1**-T**6** (FIGS. **1**B-**1**C) of first barrier metal layers **129**N**1**-**129**N**3** and **129**P**1**-**129**P**3** can be achieved by selectively forming first barrier metal layers **129***a*-**129***c* of different thicknesses in NFETs **102**N**1**-**102**N**3** and PFETs **102**P**1**-**102**P**3**, as shown in FIGS. **11**A-**11**B. The selective formation of first barrier metal layers **129***a*-**129***c* can include sequential operations of (i) blanket depositing a metal nitride layer (not shown) of thickness T**1** on the structures of FIGS. **5**A-**5**B, (ii) patterning the metal nitride layer of thickness T**1** to selectively form first barrier metal layers **129***a* on NFET **102**N**1** and PFET **102**P**1** as shown in FIGS. **11**A-**11**B, (iii) blanket depositing a metal nitride layer (not shown) of thickness T**2** on the structures formed after the formation of first barrier metal layers **129***a*, (iv) patterning the metal nitride layer of thickness T**2** to selectively form first barrier metal layers **129***b* on NFET **102**N**2** and PFET **102**P**2** as shown in FIGS. **11**A-**11**B, (v) blanket depositing a metal nitride layer (not shown) of thickness T**3** on the structures formed after the formation of first barrier metal layers **129***b*, and (vi) patterning the metal nitride layer of thickness T**3** to selectively form first barrier metal layers **129***c* on NFET **102**N**3** and PFET **102**P**3** as shown in FIGS. **11**A-**11**B. Even though first barrier metal layer **129***c* is described to be formed after first barrier metal layers **129***b*, which is described to be formed after first barrier metal layers **129***a*, first barrier metal layers **129***a*-**129***c* can be formed in any order.
(64) The blanket deposition and patterning of the metal nitride layers for first barrier metal layers **129***a*-**129***c* of FIGS. **11**A-**11**B can be similar to the blanket deposition and patterning processes of metal nitride layer **129***a**.
(65) Referring to FIG. **2**, in operation **225**, n- and p-type WFM oxide layers are selectively formed on the first barrier metal layers of NFETs and PFETs. For example, as shown in FIGS. **12**A-**13**B, nWFM oxide layer **132**N can be selectively formed on the structure of FIG. **10**A and pWFM oxide layer **132**P can be selectively formed on the structure of FIG. **10**B. Similarly, nWFM oxide layer **132**N can be selectively formed on the structure of FIG. **11**A and pWFM oxide layer **132**P can be selectively formed on the structure of FIG. **11**B, which are not shown here. During subsequent processing, nWFM oxide layer **132**N can form nWFM oxide layers **132**N**1**-**132**N**3** and pWFM oxide layer **132**P can form pWFM oxide layers **132**P**1**-**132**P**3**.
(66) Referring to FIGS. **12**A-**13**B, the selective formation of nWFM oxide layer **132**N and pWFM oxide layer **132**P can include sequential operations of (i) blanket depositing a metal oxide layer (not shown) for nWFM oxide layer **132**N on the structures of FIGS. **10**A-**10**B, (ii) patterning the metal oxide layer to selectively form nWFM oxide layer **132**N on NFETs **102**N**1**-**102**N**3** as shown in FIG. **12**A, (iii) blanket depositing a metal oxide layer (not shown) for pWFM oxide layer **132**P on the structures of FIGS. **12**A-**12**B, and (iv) patterning the metal oxide layer to selectively form pWFM oxide layer **132**P on PFETs **102**P**1**-**102**P**3** as shown in FIG. **13**B.
(67) In some embodiments, the metal oxide layer for nWFM oxide layer **132**N can include (i) REM

oxides, such as La.sub.2O.sub.3, CeO.sub.2, Yb.sub.2O.sub.3, Lu.sub.2O.sub.3, and Er.sub.2O.sub.3; (ii) oxides of a metal from group IIA (e.g., MgO or SrO), group IIIB (e.g., Y.sub.2O.sub.3), group IVB (e.g., HfO.sub.2 or ZrO.sub.2), or group VB (e.g., Ta.sub.2O.sub.5) of the periodic table; (iii) SiO.sub.2; or (iv) a combination thereof. In some embodiments, the metal oxide layer for pWFM oxide layer **132**P can include (i) Al-based metal oxides, such as Al.sub.2O.sub.3 and Al.sub.2TiO.sub.5; (ii) oxides of a metal from group VB (e.g., NbO), group IIA (e.g., B.sub.2O.sub.3), group VA (e.g., P.sub.2O.sub.5) of the periodic table; or (iii) a combination thereof.
(68) The blanket deposition of the metal oxide layer for nWFM oxide layer **132**N can include blanket depositing about 0.01 nm to about 2 nm thick metal oxide layer with an ALD or a CVD process using lanthanum tris(formamidinate) (La(FAMD).sub.3) or lanthanum tetramethylheptanedionate (La(thd).sub.3) and O.sub.3 as precursors at a temperature ranging from about 250° C. to about 350° C. In some embodiments, the metal oxide layer for nWFM oxide layer **132**N can be deposited in an ALD process of about 20 cycles to about 30 cycles, where one cycle can include sequential periods of: (i) first precursor gas (e.g., La(FAMD).sub.3 or La(thd).sub.3) flow, (ii) a first gas purging process, (iii) a second precursor gas (e.g., O.sub.3) gas flow, and (iv) a second gas purging process. The blanket deposited metal oxide layer can be substantially conformally deposited (e.g., step coverage of about 99%) on the structures of FIGS. **10**A-**10**B.
(69) The patterning of the blanket deposited metal oxide layer for nWFM oxide layer **132**N can include (i) selectively forming a masking layer (e.g., a photoresist layer or a nitride layer; not shown) on the portion of the blanket deposited metal oxide on NFETs **102**N**1**-**102**N**3**, (ii) selectively removing portions of the blanket deposited metal oxide layer on PFETs **102**P**1**-**102**P**3** to form the structures of FIGS. **12**A-**12**B, and (iii) removing the masking layer. The selectively removing can include an acid-based (e.g., HCl-based) wet or dry etching.
(70) During the blanket deposition of the metal oxide layer for nWFM oxide layer **132**N, top surface of first barrier metal layer **129***c* can be oxidized to form barrier metal oxide layers **130**N-**130**P as shown in FIGS. **12**A-**12**B. Barrier metal oxide layers **130**N-**130**P can have similar thickness. Barrier metal oxide layer **130**N can form barrier metal oxide layers **130**N**1**-**130**N**3** in subsequent processing. In some embodiments, barrier metal oxide layer **130**P can be etched during the selective removal of the portions of the blanket deposited metal oxide layer on PFETs **102**P**1**-**102**P**3** and in some embodiments, barrier metal oxide layer **130**P can remain unetched.
(71) The blanket deposition of the metal oxide layer for pWFM oxide layer **132**P can include blanket depositing about 0.01 nm to about 2 nm thick metal oxide layer with an ALD or a CVD process using trimethylaluminium (TMA) and H.sub.2O as precursors at a temperature ranging from about 250° C. to about 350° C. In some embodiments, the metal oxide layer for pWFM oxide layer **132**P can be deposited in an ALD process of about 20 cycles to about 30 cycles, where one cycle can include sequential periods of: (i) first precursor gas (e.g., TMA) flow, (ii) a first gas purging process, (iii) a second precursor gas (e.g., H.sub.2O) gas flow, and (iv) a second gas purging process. The blanket deposited metal oxide layer can be substantially conformally deposited (e.g., step coverage of about 99%) on the structures of FIGS. **12**A-**12**B.
(72) The patterning of the blanket deposited metal oxide layer for pWFM oxide layer **132**P can include (i) selectively forming a masking layer (e.g., a photoresist layer or a nitride layer; not shown) on the portion of the blanket deposited metal oxide on PFETs **102**P**1**-**102**P**3**, (ii) selectively removing portions of the blanket deposited metal oxide layer on NFETs **102**N**1**-**102**N**3** to form the structures of FIGS. **13**A-**13**B, and (iii) removing the masking layer. The selectively removing can include an acid-based (e.g., HF-based) wet or dry etching.
(73) During the blanket deposition of the metal oxide layer for pWFM oxide layer **132**P, first barrier metal layer **129***c* on PFETs **102**P**1**-**102**P**3** can be further oxidized to form barrier metal oxide layer **130**P* (FIG. **13**B), which is thicker than barrier metal oxide layer **130**P.
(74) Following the formation of pWFM oxide layer **132**P, a drive-in anneal process can be

performed on the structures of FIGS. **13**A-**13**B. The drive-in anneal process increases the metal ion (e.g., La ions and/or Al ions) concentration at the interface between nWFM oxide layers **132**N and barrier metal oxide layers **130**N and at the interface between pWFM oxide layers **132**P and barrier metal oxide layers **130**P. Increasing the metal ion concentration can increase the dipole concentration in dipole layers **131**N-**131**P (FIGS. **13**A-**13**B) induced by nWFM oxide layer **132**N and pWFM oxide layer **132**P. The above discussion of dipole layers **131**N**1**-**131**N**3** applies to dipole layer **131**N. The above discussion of dipole layers **131**P**1**-**131**P**3** applies to dipole layer **131**P.

(75) The drive-in anneal process can include annealing nWFM oxide layers **132**N and pWFM oxide layer **132**P at a temperature from about 550° C. to about 850° C. and at a pressure from about 1 torr to about 30 torr for a time period ranging from about 0.1 sec to about 30 sec. In some embodiments, the drive-in anneal process can include two anneal processes: (i) a soak anneal process at a temperature from about 550° C. to about 850° C. for a time period ranging from about 2 sec to about 60 sec and (ii) a spike anneal process at a temperature from about 700° C. to about 900° C. for a time period ranging from about 0.1 sec to about 2 sec.

(76) Referring to FIG. **2**, in operation **230**, second barrier metal layers, FFW layers, and gate metal fill layers are formed on the n- and p-type WFM oxide layers. For example, as shown in FIGS. **14**A-**14**B, second barrier metal layers **133**, FFW layers **134**, and gate metal fill layers **135** can be formed on the structures of FIGS. **13**A-**13**B. The material for second barrier metal layers **133** can be blanket deposited on the structures of FIGS. **13**A-**13**B. The material FFW layers **134** can be blanket deposited on the material for second barrier metal layers **133**. The material for gate metal fill layers **135** can be blanket deposited on the material for FFW layers **134**. Following these blanket depositions, HK gate dielectric layer **128**, first barrier metal layers **129***a*-**129***c*, barrier metal oxide layers **130**N-**130**P, nWFM oxide layer **132**N, pWFM oxide layer **132**P, the material for second barrier metal layers **133**, the material for FFW layers **134**, and the material for gate metal fill layers **135** can be polished by a chemical mechanical polishing process to form the structures of FIGS. **14**A-**14**B. Thus, as described in operations **215**-**230**, gate structures **112**N**1**-**112**N**3** and **112**P**1**-**112**P**3** can be formed with at least three different threshold voltages on the same substrate **106**.

(77) The present disclosure provides example FET gate structures with different EWF values to form FETs (e.g., GAA FETs and/or finFETs) with different threshold voltages and provides example methods of forming such FETs on the same substrate. The example methods form NFETs NFETs (e.g., NFETs **102**N**1**-**102**N**3**) and PFETs PFETs (e.g, PFETs **102**P**1**-**102**P**3**) with different gate structure configurations (e.g., gate structures **112**N**1**-**112**N**3** and **112**P**1**-**112**P**3**), but with similar WFM layer thicknesses can be selectively formed on the same substrate (e.g., substrate **106**) to achieve threshold voltages (e.g., threshold voltages V**1**-V**6**) different from each other. The different gate structure configurations can have barrier metal layers (e.g., barrier metal layers **129**N**1**-**129**N**3** and **129**P**1**-**129**P**3**) of different thicknesses disposed between the WFM layers and high-K gate dielectric layers (e.g., HK gate dielectric layers **128**N**1**-**128**N**3** and **128**P**1**-**128**P**3**). In addition, the WFM layers can include WFM oxide layers (e.g., WFM oxide layers **132**N**1**-**132**N**3** and **132**P**1**-**132**P**3**) that induce dipole layers (e.g., dipole layers **131**N**1**-**131**N**3** and **131**P**1**-**131**P**3**) at the interface between the WFM layers and the barrier metal layers. The different barrier metal layer thicknesses (e.g., thicknesses T**1**-T**6**) provide different spacings (e.g., spacings S**1**-S**6**) between the WFM layers and high-K gate dielectric layers and different spacings (e.g., spacings S**7**-S**12**) between the induced dipole layers and the high-K gate dielectric layers. These different spacings result in the FET gate structures having EWF values (e.g., EWF values E**1**-E**6**) different from each other and consequently having threshold voltages different from each other. Thus, tuning the barrier metal layer thicknesses can tune the EWF values of the NFET and PFET gate structures and, as a result, adjust the threshold voltages of the NFETs and PFETs without varying their WFM layer thicknesses.

(78) In some embodiments, a semiconductor device includes a substrate, first and second pair of epitaxial source/drain (S/D) regions disposed on the substrate, first and second nanostructured channel regions disposed between the epitaxial S/D regions of the first pair of epitaxial S/D regions and between the epitaxial S/D regions of the second pair of epitaxial S/D regions, respectively, and first and second gate structures with effective work function values different from each other. The first and second gate structures include first and second high-K gate dielectric layers surrounding the first and second nanostructured channel regions, respectively, first and second barrier metal layers with thicknesses different from each other disposed on the first and second high-K gate dielectric layers, respectively, first and second work function metal (WFM) oxide layers with thicknesses substantially equal to each other disposed on the first and second barrier metal layers, respectively, a first dipole layer disposed between the first WFM oxide layer and the first barrier metal layer, and a second dipole layer disposed between the second WFM oxide layer and the second barrier metal layer.
(79) In some embodiments, a semiconductor device includes a substrate, first and second pair of epitaxial source/drain (S/D) regions disposed on the substrate, first and second fin regions disposed between the epitaxial S/D regions of the first pair of epitaxial S/D regions and between the epitaxial S/D regions of the second pair of epitaxial S/D regions, respectively. The semiconductor device further includes first and second gate structures having first and second high-K gate dielectric layers disposed on the first and second fin regions, respectively, first and second barrier metal layers with thicknesses different from each other disposed on the first and second high-K gate dielectric layers, respectively, first and second barrier metal oxide layers disposed on the first and second barrier metal layers, respectively, an aluminum (Al)-free rare-earth metal-based (REM-based) work function metal (WFM) oxide layer disposed on the first barrier metal oxide layer, and an Al-based WFM oxide layer disposed on the second barrier metal oxide layer.
(80) In some embodiments, a method includes forming first and second nanostructured channel regions in a fin structure on a substrate, forming first and second high-K gate dielectric layers surrounding the first and second nanostructured channel regions, respectively, forming first and second barrier metal layers of different thicknesses on the first and second high-K gate dielectric layers, forming first and second work function metal (WFM) oxide layers of substantially equal thicknesses on the first and second barrier metal layers, respectively, performing a drive-in anneal process on the first and second WFM oxide layers, forming third and fourth barrier metal layers of substantially equal thicknesses on the first and second WFM oxide layers, and forming first and second gate metal fill layers on the third and fourth barrier metal layers, respectively.
(81) The foregoing disclosure outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

## Claims

1. A semiconductor device, comprising: a substrate; a nanostructured channel region disposed on the substrate; a gate structure, comprising: a gate dielectric layer disposed on the nanostructured channel region; a first metal nitride layer disposed on the gate dielectric layer; a first metal oxide layer disposed on the first metal nitride layer; a second metal oxide layer disposed on the first metal oxide layer; a second metal nitride layer disposed on the second metal oxide layer; and a metal layer disposed on the second metal nitride layer.

2. The semiconductor device of claim 1, further comprising a dipole layer disposed between the first and second metal oxide layers.
3. The semiconductor device of claim 1, further comprising a fluorine-free metal layer disposed between the second metal nitride layer and the metal layer.
4. The semiconductor device of claim 1, wherein the first metal nitride layer and the first metal oxide layer comprise a same metal.
5. The semiconductor device of claim 1, wherein the first and second metal nitride layers comprise a same metal.
6. The semiconductor device of claim 1, wherein the second metal oxide layer comprises an aluminum-free metal oxide.
7. The semiconductor device of claim 1, wherein the second metal oxide layer comprises an oxide of a rare-earth metal.
8. The semiconductor device of claim 1, wherein the gate structure further comprises a concentration profile of nitrogen atoms across the first metal nitride layer and the first metal oxide layer, and wherein a peak of the concentration profile is at an interface between the first metal nitride layer and the first metal oxide layer.
9. The semiconductor device of claim 1, wherein the gate structure further comprises a concentration profile of metal atoms across the first and second metal oxide layers, and wherein a peak of the concentration profile is at an interface between the first and second metal oxide layers.
10. The semiconductor device of claim 1, wherein the first metal oxide layer comprises lanthanum oxide.
11. A semiconductor device, comprising: a substrate; first and second nanostructured layers disposed on the substrate; a first gate structure, comprising: a first metal nitride layer surrounding the first nanostructured layer; and a first metal oxide layer disposed on the first metal nitride layer; and a second gate structure, comprising: a second metal nitride layer surrounding the second nanostructured layer; and a second metal oxide layer disposed on the second metal nitride layer, wherein the first and second metal nitride layers have thicknesses different from each other, and wherein the first and second metal oxide layers have thicknesses substantially equal to each other.
12. The semiconductor device of claim 11, wherein the first metal nitride layer and the first metal oxide layer comprise a same metal.
13. The semiconductor device of claim 11, wherein the second metal nitride layer and the second metal oxide layer comprise a same metal.
14. The semiconductor device of claim 11, wherein the first gate structure further comprises a rare-earth metal-based oxide layer disposed on the first metal oxide layer.
15. The semiconductor device of claim 14, further comprising a dipole layer disposed between the rare-earth metal-based oxide layer and the first metal oxide layer.
16. The semiconductor device of claim 11, wherein the second gate structure further comprises an aluminum-based oxide layer disposed on the second metal oxide layer.
17. A method, comprising: depositing a gate dielectric layer on a fin structure; depositing a first metal nitride layer on the gate dielectric layer; depositing a first metal oxide layer on the first metal nitride layer to convert a top portion of the first metal nitride layer into a second metal oxide layer; performing a drive-in anneal process on the second metal oxide layer; depositing a second metal nitride layer on the first metal oxide layer; and depositing a gate metal fill layer on the second metal nitride layer.
18. The method of claim 17, wherein depositing the first metal oxide layer comprises depositing a rare-earth metal-based oxide layer on the first metal nitride layer.
19. The method of claim 17, wherein depositing the first metal oxide layer comprises depositing an aluminum-based oxide layer on the first metal nitride layer.
20. The method of claim 17, wherein depositing the first and second metal nitride layers comprises depositing titanium nitride layers.